Measurement, Error, and Significant Figures — Think Before You Round

Mini-Review (read first)

Precision vs. Accuracy
• **Precision** = repeatability/consistency (small spread).
• **Accuracy** = closeness to the true value.

Approximate Error
• Absolute error: $|\,\text{measured} - \text{true}\,|$.
• Relative error: $\dfrac{\text{abs. error}}{\text{true}}$.
• Percent error: relative error $\times 100\%$.

Rounding & Intervals
• If a value is rounded to the nearest u (unit, tenth, hundredth, etc.), the *true* value lies in $\left[v - \frac{u}{2},\, v + \frac{u}{2}\right)$.
• Propagate ranges through arithmetic to bound results.

Significant Figures (Sig Figs)
Counting rules:
• Nonzero digits are significant.
• Zeros between sig digits are significant.
• Leading zeros are *not* significant.
• Trailing zeros are significant *if* a decimal point is shown; otherwise ambiguous (use scientific notation).

Ops rules:
• **Add/Subtract:** round *to the least precise decimal place* among operands.
• **Multiply/Divide:** round *to the least number of sig figs* among operands.

Scientific notation: $a \times 10^n$ with $1 \le |a| < 10$ carries the sig figs of a.

1. (**Precision vs accuracy**) A thermometer reads $20.1℃, 20.0℃, 20.1℃, 20.0℃$ for a sample known to be $19.9℃$. Comment on precision and accuracy.

2. (**Absolute & percent error**) A metal rod is marked as 50.0 cm (true length), but a ruler reads 49.6 cm. Find absolute error and percent error.

3. (**Interval from rounding**) A measurement recorded as 7.84 m is stated to be rounded to the nearest 0.01 m. Give the interval where the true value lies.

4. (**Backwards rounding**) A student rounded a length to 3.4 cm to the nearest 0.1 cm. Give the set of all true lengths that could lead to 3.4 cm when rounded.

5. (**Range propagation — sum**) Two boards measured as 1.25 m and 2.4 m were each rounded to the nearest 0.01 m and 0.1 m respectively. Give the possible interval for the true total length.

6. (**Range propagation — product**) A rectangle has measured length $L = 4.20$ m (nearest 0.01 m) and width $W = 1.8$ m (nearest 0.1 m). Give a *range* for the true area by interval multiplication.

7. (**Sig fig counting**) For each, state the number of significant figures: (a) 0.00430 (b) 1200 (c) 1.200×10^3 (d) 300.

8. (**Add/sub with sig figs**) Compute $(12.07 + 0.3 + 0.025)$ using the proper decimal place rule.

9. (**Mult/div with sig figs**) Compute (3.20)(0.054) with correct sig figs and scientific

notation.

10. (**Mixed ops carefully**) Evaluate $\dfrac{(7.16 - 3.2)}{0.041}$ applying add/sub rule first, then mult/-div rule.

11. (**Choose a better instrument**) Two balances: A reads to 0.1 g, B to 0.01 g. You need *high precision but unknown true mass.* Which and why? How does this affect percent error if true mass is about 25 g?

12. (**Accuracy trap**) Five measurements: $9.98, 9.99, 9.98, 9.99, 9.98$ cm. The standard is 10.50 cm. Explain precision and accuracy; which is high/low?

13. (**Bounding a circumference**) A circular lid's radius is measured as $r = 12.4$ cm to the nearest 0.1 cm. Give the interval for the true radius and then bound the *true* circumference $C = 2\pi r$.

14. (**Area with dependent rounding**) A square's side is recorded as $s = 3.6$ cm (nearest 0.1 cm). Give the range for the true area $A = s^2$.

15. (**Percent error from rounding**) If a distance rounded to the nearest meter is reported as $d = 142$ m, what is the *maximum* possible percent error?

16. (**Sig figs and zeros**) Rewrite each with 3 significant figures using scientific notation: (a) 0.0004571 (b) 6020000 (c) 300

17. (**When to round**) Calculate the area of a rectangle from 9.4 cm (nearest 0.1) by 2.15 cm (nearest 0.01). Show the unrounded product first; then report with proper sig figs; then give a range using intervals.

18. (**Targeting a decimal place**) You add 3.418 m (nearest 0.001) and 0.27 m (nearest 0.01). To what decimal place must the sum be rounded, and what is the final reported value?

19. (**Relative vs absolute error**) Two lengths: $A = 100.0$ cm and $B = 10.00$ cm, both measured to the nearest 0.1 cm and 0.01 cm respectively. Which has the larger *relative* uncertainty? Explain numerically.

20. (**Division with range**) Density $\rho = \dfrac{m}{V}$. A mass $m = 12.0$ g (nearest 0.1 g), volume $V = 4.0\ \text{cm}^3$ (nearest 0.1 cm^3). Give (i) a sig-fig answer and (ii) a range for ρ using intervals.

21. (**Nonlinear propagation idea**) If $A = \pi r^2$ with $r = 8.0$ cm (nearest 0.1), which r in the radius interval gives the largest A? Bound A.

22. (**Hidden precision**) A digital display shows 2.40 V. Explain the implied precision and the possible interval for the true voltage, then state the number of sig figs.

23. (**Compound calculation**) Compute $\dfrac{(1.205 + 0.88)}{(0.0315)}$ with correct rules; report in scien-

tific notation with the proper sig figs.

24. (**Rounding order matters**) Which is more correct under sig-fig rules: rounding intermediate steps or only the final result? Demonstrate with $(4.37 + 1.2) \times 0.015$.

25. (**Concept synthesis**) A length x is measured as 5.20 cm (nearest 0.01), a width y as 1.3 cm (nearest 0.1), and a height z as 0.84 cm (nearest 0.01). (a) Give intervals for x, y, z. (b) Find a plausible range for volume $V = xyz$. (c) Report V once using sig-fig rules and once using your interval bounds (as an inequality).